

Suffix-Tree Example: minimize

1 Text and Suffixes

Text

The example text is

$T = \text{minimize.}$

Its suffixes are:

Position	Suffix	With end marker
1	minimize	minimize\$
2	inimize	inimize\$
3	nimize	nimize\$
4	imize	imize\$
5	mize	mize\$
6	ize	ize\$
7	ze	ze\$
8	e	e\$

The position is important because a suffix-tree leaf can store where that suffix begins in the original text.

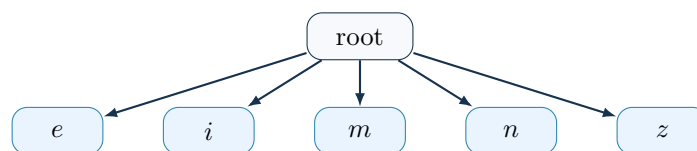
2 Root Branches

The first characters of the suffixes are

$m, i, n, i, m, i, z, e.$

Therefore, the root has branches beginning with

$e, i, m, n, z.$

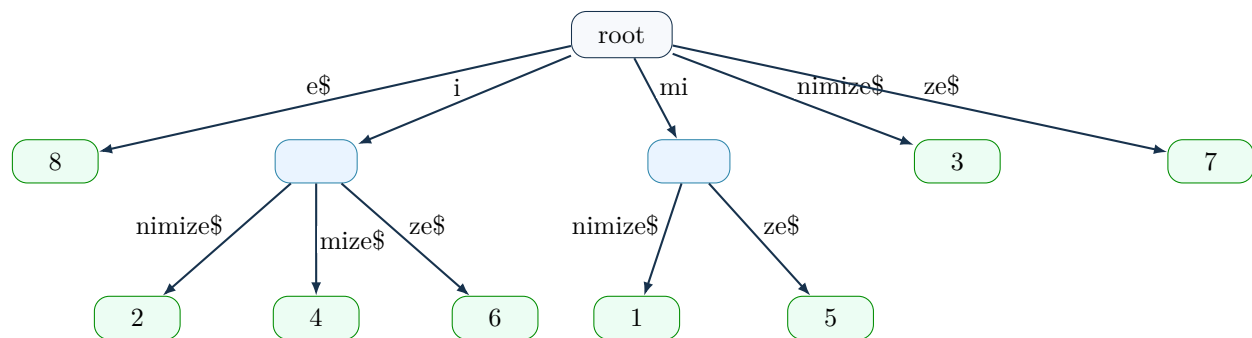


repeated first letters share one branch

3 Grouped Suffixes

Root branch	Suffixes in that group, with starting positions
<i>e</i>	8 : e\$
<i>i</i>	2 : imimize\$, 4 : imize\$, 6 : ize\$
<i>m</i>	1 : minimize\$, 5 : mize\$
<i>n</i>	3 : nimize\$
<i>z</i>	7 : ze\$

4 Full Suffix Tree



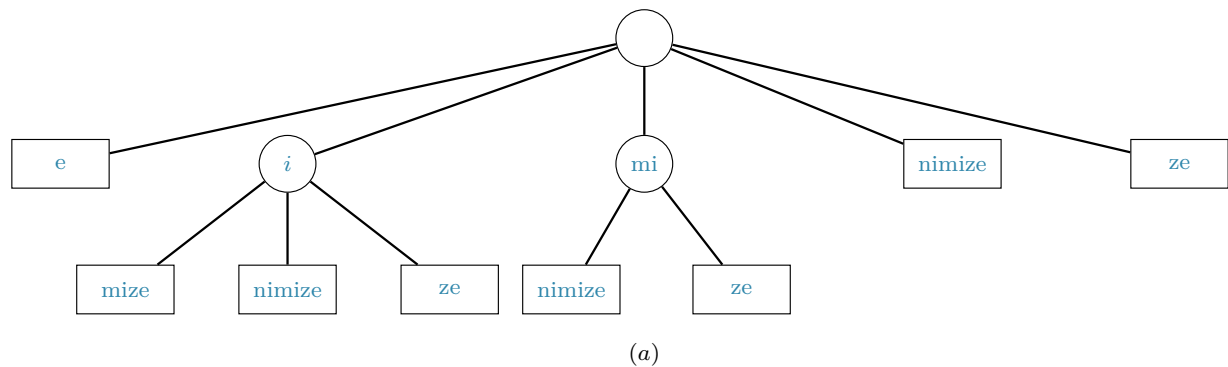
How to read the tree

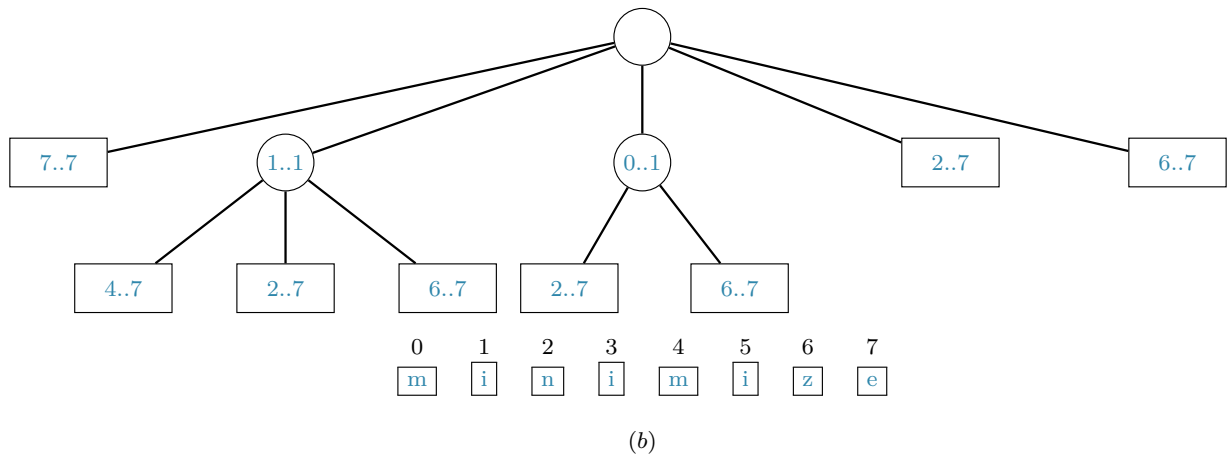
Each leaf stores the starting position of one suffix. The full suffix is obtained by reading edge labels from the root to that leaf.

5 String-Label and Index-Range Views

The same suffix tree can be drawn in two equivalent ways:

- using actual edge labels such as *nimize* and *ze*;
- using index ranges $j..k$ into the original text $T = \text{minimize}$.





Index-Range Meaning

An edge label $j..k$ means the edge stores the substring $T[j..k]$. For example, $2..7$ represents minimize, and $6..7$ represents ze.

6 Compression Idea

The suffix tree stores shared prefixes of suffixes once, and then compresses nonbranching chains into substring-labelled edges.

Interpretation

Branches such as i , m , and ze show the same compression principle used in compressed tries: common prefixes are shared, and one-child chains become longer edge labels.

7 Worked Pattern Queries

Query 1

Search for

$P = \text{mi.}$

Solution

The pattern starts with m , so search begins on the m -branch. The suffixes under that branch are

1 : minimize\$, 5 : mize\$.

Both begin with mi , so the pattern occurs at positions

1 and 5.

Query 2

Search for

$P = \text{ize.}$

Solution

The pattern starts with i , so search begins on the i -branch. Among the grouped suffixes,

6 : ize\$

starts with ize. Therefore,

ize occurs at position 6.
